Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Fall semester 2025-2027

MAVLD504 – ANALOG IC DESIGN

SLOT: L3 + L4

Submitted to

SENSE

School of Electronics Engineering

EXPERIMENT 6

Submitted by

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Design of Two Stage Operational Amplifier

Aim:
To design Two Stage Operational Amplifier with given specifications:
Gain (Av) = 1000 v/v or 60dB, Gain Bandwidth Product = 30MHz, Slew Rate = 20 V/µs, Input Common Mode Range = 1.6V(max) - 0.8V(min), Power Consumption < 0.3mW, CL=2pF, Phase Margin (PM) >=60 degrees.
Apparatus:
Cadence Virtuoso, UMC180nmPDK
Circuit Diagram:

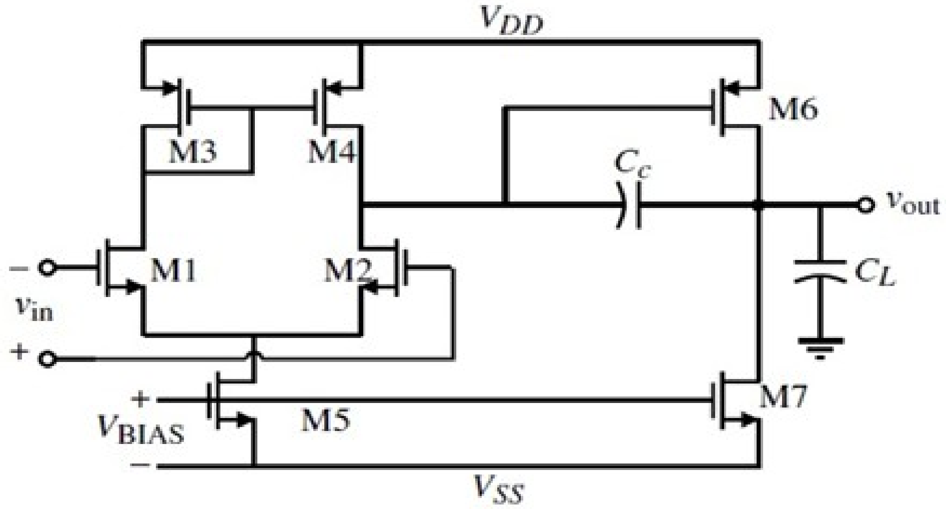


Fig 1: Circuit Diagram

Schematic Procedure:

Open Terminal and create a directory in the server using 'ssh -Y 25mvd0100@10.110.5.59'. To open the Cadence Virtuoso 6.1.8 software, give next command as 'csh'. Now give command as 'virtuoso &' opens the popup of virtuoso software.

To add the required components on schematic, go to create an instance and choose the lib as UMC_18_CMOS and select required N_18_MM (nmos) and P_18_MM (pmos) and place it in schematic. In instance choose 'analoglib' for voltage sources(vdc), ground (gnd), cap (capacitor) and vin (AC Voltage source).

After placing the required components on the schematic connect the connections using required wires as in circuit diagram. Now by selecting the component and pressing Q give values. For both nmos and pmos give length as 1µm, Capacitor is 10pF. For AC Voltage Source give Amplitude is 10µV, Frequency is 1kHz, AC Magnitude is 1 and AC Phase is 0 and other voltage source give AC Phase as 180. For DC voltages give Vdd as 1.8V. Now save the circuit.
Transfer Function:
The approximate Transfer Function is written as

$$TF = \frac{V_{out}}{V_{in}} = \frac{-g_m R_D}{(1+\frac{s}{w_{in}})(1+\frac{s}{w_{out}})}$$

$$w_{in} = \frac{1}{((1 + g_m R_D)C_{gd} + C_{gs})R_S}$$
$$w_{out} = \frac{1}{((1 + \frac{1}{g_m R_D})C_{gd} + C_{db} + C_L)R_D}$$

Calculations:

- **From UMC_180 Spice parameters**,

  NMOS

  $$\text{Vth=0.3V, µn} = 3.14 * 10^{-2}\frac{m^2}{VS}, \text{Cox} = \frac{\text{£ox}}{\text{tox}} = \frac{3.9*8.854*10^{-12}}{4.2*10^{-9}} = 8.221 * 10^{-3}\frac{F}{cm^2}$$

  $$\mu_n C_{ox} = 258\ \frac{\mu A}{V^2}$$

  PMOS

  $$\text{Vth=-0.45V, µp} = 1.145 * 10^{-2}\frac{m^2}{VS}, \text{Cox} = \frac{\text{£ox}}{\text{tox}} = \frac{3.9*8.854*10^{-12}}{4.2*10^{-9}} = 8.221 * 10^{-3}\frac{F}{cm^2}$$

  $$\mu_p C_{ox} = 94\ \frac{\mu A}{V^2}$$

- $C_C \geq 0.22C_L \geq 0.22(2p) \geq 0.44pF \Rightarrow Assume\ C_C = 0.8pF$
- **Gain Bandwidth** $GB = \frac{g_{m1}}{2*3.14*C_C} \Rightarrow g_{m1} = GB * 2 * 3.14 * C_C = 151\mu S$
- **Slew Rate** SR $= \frac{dV_{out}}{dt} = \frac{I_5}{C_C} \Rightarrow \frac{20V}{\mu s} = \frac{I5}{0.8pF} \Rightarrow I5 = 16\mu A$
- $g_m = \sqrt{2I_D \mu_n C_{ox}\left(\frac{W}{L}\right)} \Rightarrow \left(\frac{W}{L}\right)1 = 5.52$

Design of M3, M4

- **Input Common-Mode Range (ICMR)**
  - **ICMR (Max) Condition**: The max condition for M1 to be in saturation.

$$V_{DS(M1)} \geq V_{GS(M1)} - V_{th1} \Rightarrow V_D \geq V_G - V_{th1}$$
$$V_{DD} - V_{SG3} \geq V_{ICMR(MAX)} - V_{th1}$$
$$Hence\ V_{ICMR(MAX)} = V_{DD} - V_{SG3} + V_{th1}$$
$$1.6 \leq 1.8 - VsgM3 + 0.45$$
$$\text{Vsg3<=0.65}$$

$$I_{D3,4} = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)_{3,4}(V_{GS} - V_{TH})^2 \rightarrow \left(\frac{W}{L}\right)_{3,4} = 4.255$$

Design of M6, M7

PM>=60 deg.

$$g_{m6} \geq 10g_{m1} \geq 10 * 151\mu = 1510\mu S$$

To achieve proper mirror from 1st stage to 2nd stage.

$$\frac{\left(\frac{W}{L}\right)_6}{\left(\frac{W}{L}\right)_4} = \frac{g_{m6}}{g_{m4}} \rightarrow$$

$$\left(\frac{W}{L}\right)_6 = \left(\frac{W}{L}\right)_4 * \frac{g_{m6}}{g_{m4}} = 4.255 * \frac{1510}{79.96} = 80.21$$

- $g_{m4} = \sqrt{2I_{D4}\mu_p C_{ox}\left(\frac{W}{L}\right)}$ ➔ On solving $g_{m4} = 79.96\mu S$

Using same current I6 aspect ratio of M7 is

$$I6 = I7 = \frac{\left(\frac{W}{L}\right)_6}{\left(\frac{W}{L}\right)_4} I4 \rightarrow I7 = 144\mu A$$

Mirroring M5 & M7

$I_{D5} = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)(V_{DSat5})^2 ==> \left(\frac{W}{L}\right)_5$ =400n

$V_{DSat5} \leq V_{in,min} - V_{gsM1} \leq 0.195V \rightarrow V_{DSat5} = 0.195V$

$\left(\frac{W}{L}\right)_7 = \frac{I7}{I5}\left(\frac{W}{L}\right)_5 \rightarrow \left(\frac{W}{L}\right)_7$ =810n

Final Transistor Dimensions

Transistor	**W/L**	**Width (μm) at L=1 μm**
M1,2	7	7
M3,4	68	68
M6	80.21	80.21
M7	810n	810n
M5,8	400n	400n

Schematic Diagram:

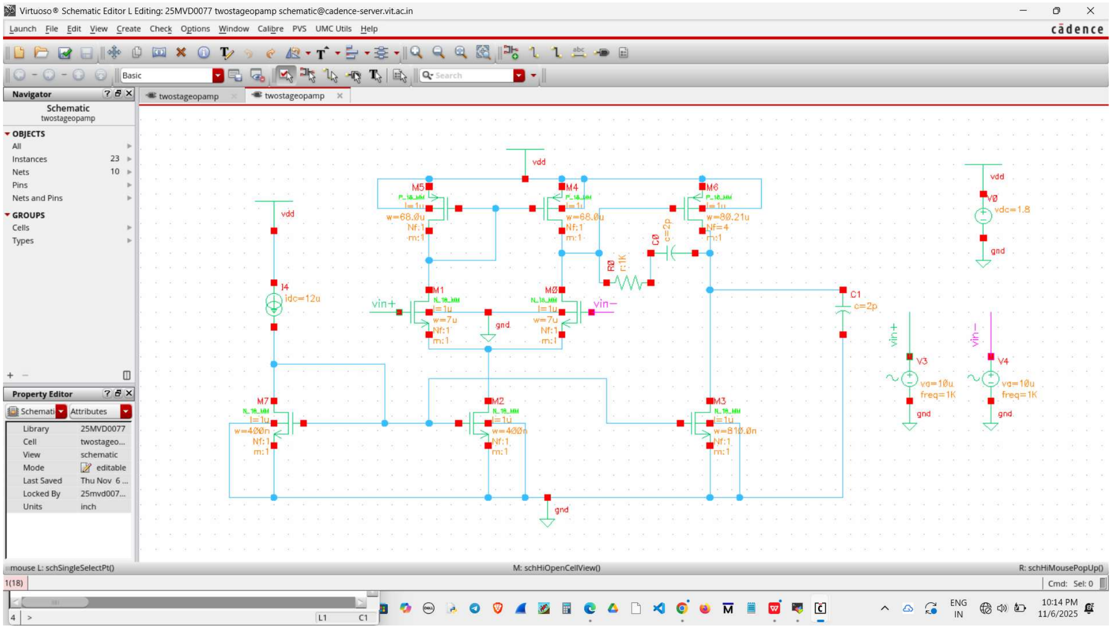


Fig 2: Schematic of Two stage opamp

Analysis Procedure:

Launch ADL:

- For DC Analysis ➔
  Analyse → Select dc → Save Operating point → Save.
- For Transient Analysis ➔
  Analyse → Select tran → Give value as 5mS → Select Conservative → Save.
- For AC Analysis ➔
  Analyse → Select Frequency → Range from 1 to 10G Hz → Save.
- For Pole Zero Analysis ➔
  Analyse → Select pz → select output and input as Voltage ➔
  Positive → Select → Click the wire above the Capacitor.
  Negative → Select → Click the wire below the Capacitor.
- For Noise Analysis ➔
  Analyse → Select noise → Set start and stop time as 1-50MHz.
  select output and input as Voltage ➔
  Positive Output → Select → Click the wire above the Capacitor.
  Negative Output → Select → Click the wire below the Capacitor.
  Input Voltage Source → Select → V0.

Now save the data and simulate the schematic.

- To view Pole Zero Analysis ➔
  Pole Zero Analysis Output is displayed on Transcript Page.
- To view DC Analysis ➔
  Right Click on Schematic → Setup → Select Lib UMC_18_MM → Select cell name → select instance.
  Setup → Global → Parameters → DC Operating point.

Setup → Global → Parameters → Double click empty variables and give region and gm. → Repeat for both Pmos and Nmos Transistors.

- To View AC Analysis ➔

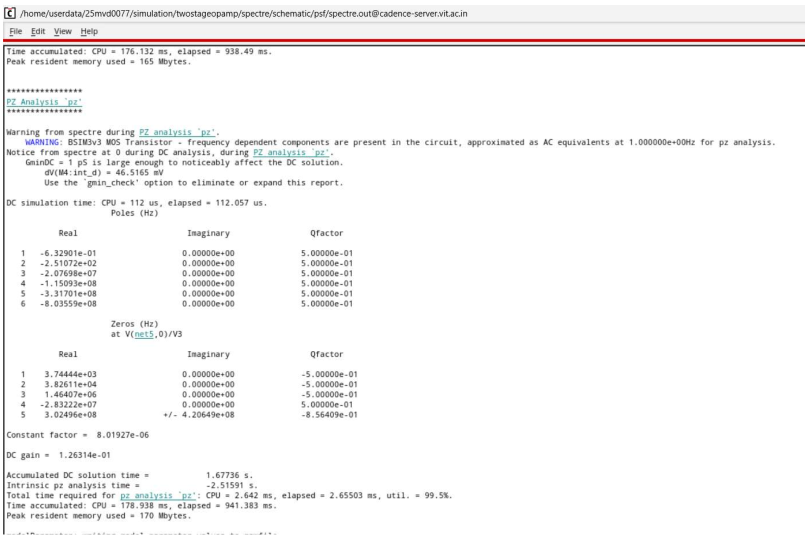
  Results → Direct Plots → AC Gain and Phase → Select the Output Node on Schematic → Select the Input Node on Schematic.
  The Gain and Phase Plot displays in new window.
- To View Transient Analysis ➔

  Results → Direct Plots → Transient minus DC → Select the Output Node on Schematic → Select the Input Node on Schematic.
- To View Noise Analysis ➔

  Results → Main Form → Select noise analysis → Output Noise as Function → Plot the Noise Plot.
  The Noise Plot displays in new window.
  Results → Print → Noise Summary → Select Integrated Noise and frequency from 1 to 5MHz. → Include all types → OK.
  The Noise Summary is displayed in new window.

Results:

The Two Stage Operational Amplifier with given specifications is designed and can be observed here.



```
/home/userdata/25mvd0077/simulation/twostageopamp/spectre/schematic/psf/spectre.out@cadence-server.vit.ac.in
File  Edit  View  Help
Time accumulated: CPU = 176.132 ms, elapsed = 938.49 ms.
Peak resident memory used = 165 Mbytes.


****************
PZ Analysis `pz'
****************

Warning from spectre during PZ analysis `pz'.
    WARNING: BSIM3v3 MOS Transistor - frequency dependent components are present in the circuit, approximated as AC equivalents at 1.000000e+00Hz for pz analysis.
Notice from spectre at 0 during DC analysis, during PZ analysis `pz'.
    GminDC = 1 pS is large enough to noticeably affect the DC solution.
        dV(M4:int_d) = 46.5165 mV
        Use the `gmin_check' option to eliminate or expand this report.

DC simulation time: CPU = 112 us, elapsed = 112.057 us.
                      Poles (Hz)

           Real                        Imaginary                 Qfactor

   1   -6.32901e-01                  0.00000e+00              5.00000e-01
   2   -2.51072e+02                  0.00000e+00              5.00000e-01
   3   -2.07698e+07                  0.00000e+00              5.00000e-01
   4   -1.15093e+08                  0.00000e+00              5.00000e-01
   5   -3.31701e+08                  0.00000e+00              5.00000e-01
   6   -8.03559e+08                  0.00000e+00              5.00000e-01

                      Zeros (Hz)
                      at V(net5,0)/V3

           Real                        Imaginary                 Qfactor

   1    3.74444e+03                  0.00000e+00             -5.00000e-01
   2    3.82611e+04                  0.00000e+00             -5.00000e-01
   3    1.46407e+06                  0.00000e+00             -5.00000e-01
   4   -2.83222e+07                  0.00000e+00              5.00000e-01
   5    3.02496e+08               +/- 4.20649e+08             -8.56409e-01

Constant factor =  8.01927e-06

DC gain =  1.26314e-01

Accumulated DC solution time =           1.67736 s.
Intrinsic pz analysis time =             -2.51591 s.
Total time required for pz analysis `pz': CPU = 2.642 ms, elapsed = 2.65503 ms, util. = 99.5%.
Time accumulated: CPU = 178.938 ms, elapsed = 941.383 ms.
Peak resident memory used = 170 Mbytes.
```

Fig 3: Pole Zero Analysis

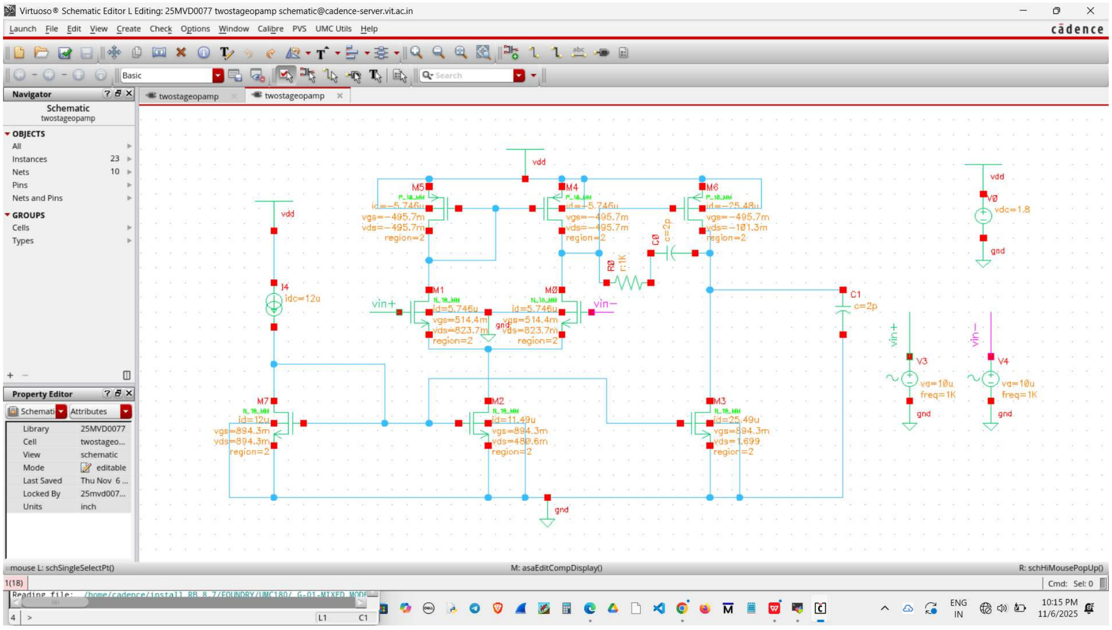


Fig 4: DC Analysis Values displaying on Schematic.

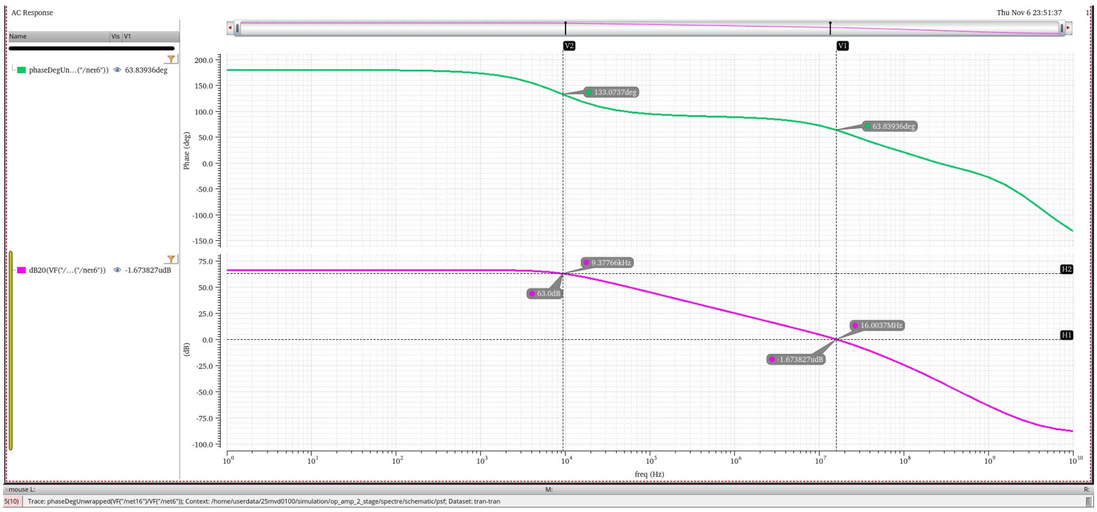


Fig 5: AC Analysis with 0dB and f3dB values.

f_{0dB} = 16.20 MHz

f_{3dB} = 09.38 kHz

Maximum gain, Av ≈ 68dB

Gain Crossover Frequency = 16.20 MHz.

Transfer Function is

$$\mathbf{H(S)} = \frac{1065.8 * (1 + \frac{S}{8.96 * 10^3})(1 + \frac{S}{3.92 * 10^7})(1 + \frac{S}{10.93 * 10^7})(1 + \frac{S}{1.87 * 10^8})(1 + \frac{S}{1.94 * 10^8})(1 + \frac{S}{5.99 * 10^8})}{(1 + \frac{S}{9.18 * 10^7})(1 + \frac{S}{9.18 * 10^7 - j6.84 * 10^6})(1 + \frac{S}{2.63 * 10^8})(1 + \frac{S}{8.38 * 10^8})(1 + \frac{S}{7.73 * 10^9})(1 + \frac{S}{1.26 * 10^{10}})}$$

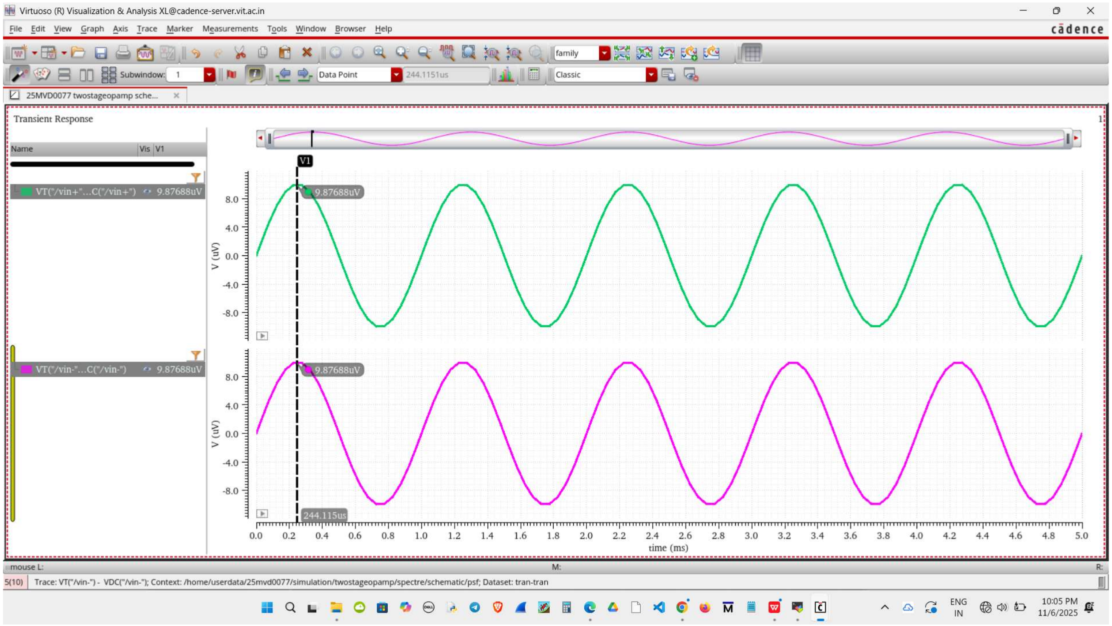


Fig 6: Transient Analysis (with Vp-p values)
Vout=41.7mV(approx.)
Vin=21µV(approx.)